

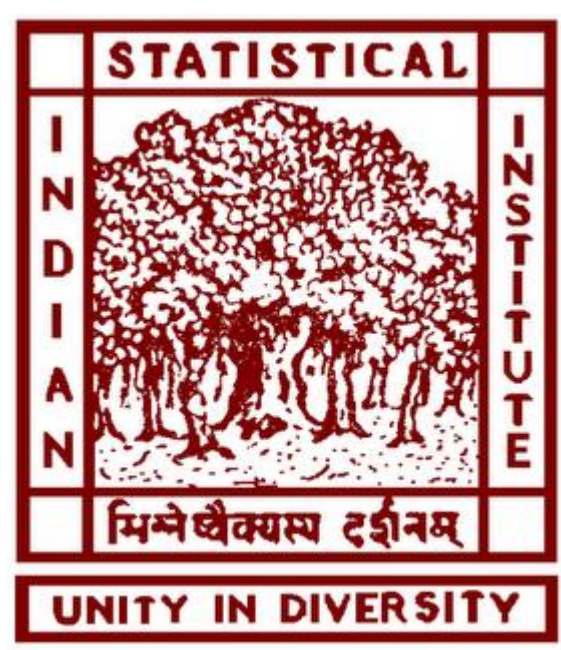
MSTIS: Multi-Views Scene Text Image Sequencing to Enhance Text Detection Performance

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MOTIVATION

- Multi-view images suffer from viewpoint variations.
- Perspective distortion, occlusion and clutter reduce text detection accuracy.
- Correct view ordering improves downstream text detection.



VIEW-1(FRONT)



VIEW-2(LEFT)

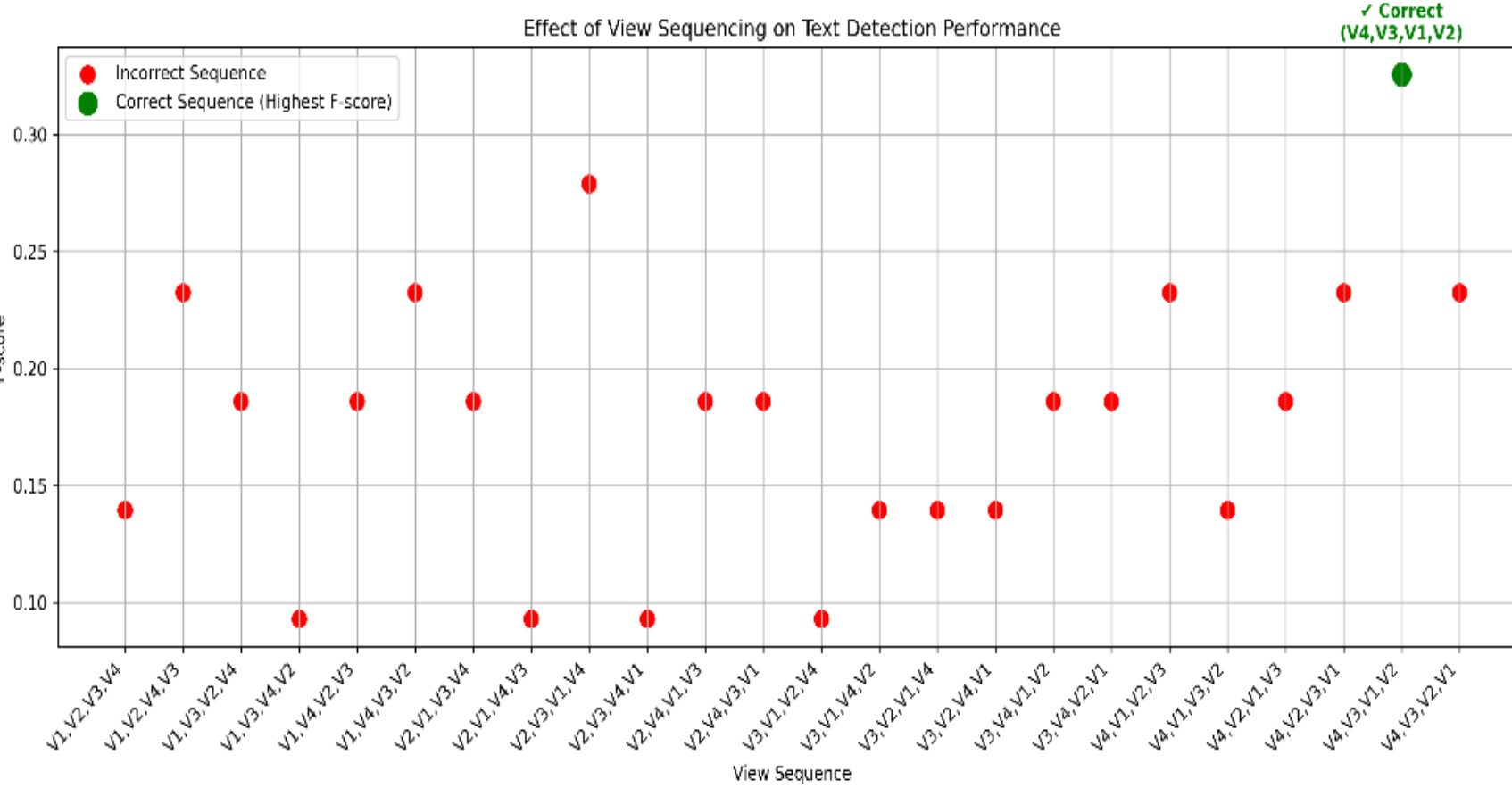


VIEW-3(RIGHT)



VIEW-4(TOP)

WHY VIEW SEQUENCING ?



Impact of View Sequencing on Text Detection Performance.

- Correct ordering consistently achieves the highest F-score.
- Incorrect ordering degrades scene text detection.

DATASET OVERVIEW



VIEW-1(FRONT)



VIEW-2(LEFT)



VIEW-3(RIGHT)



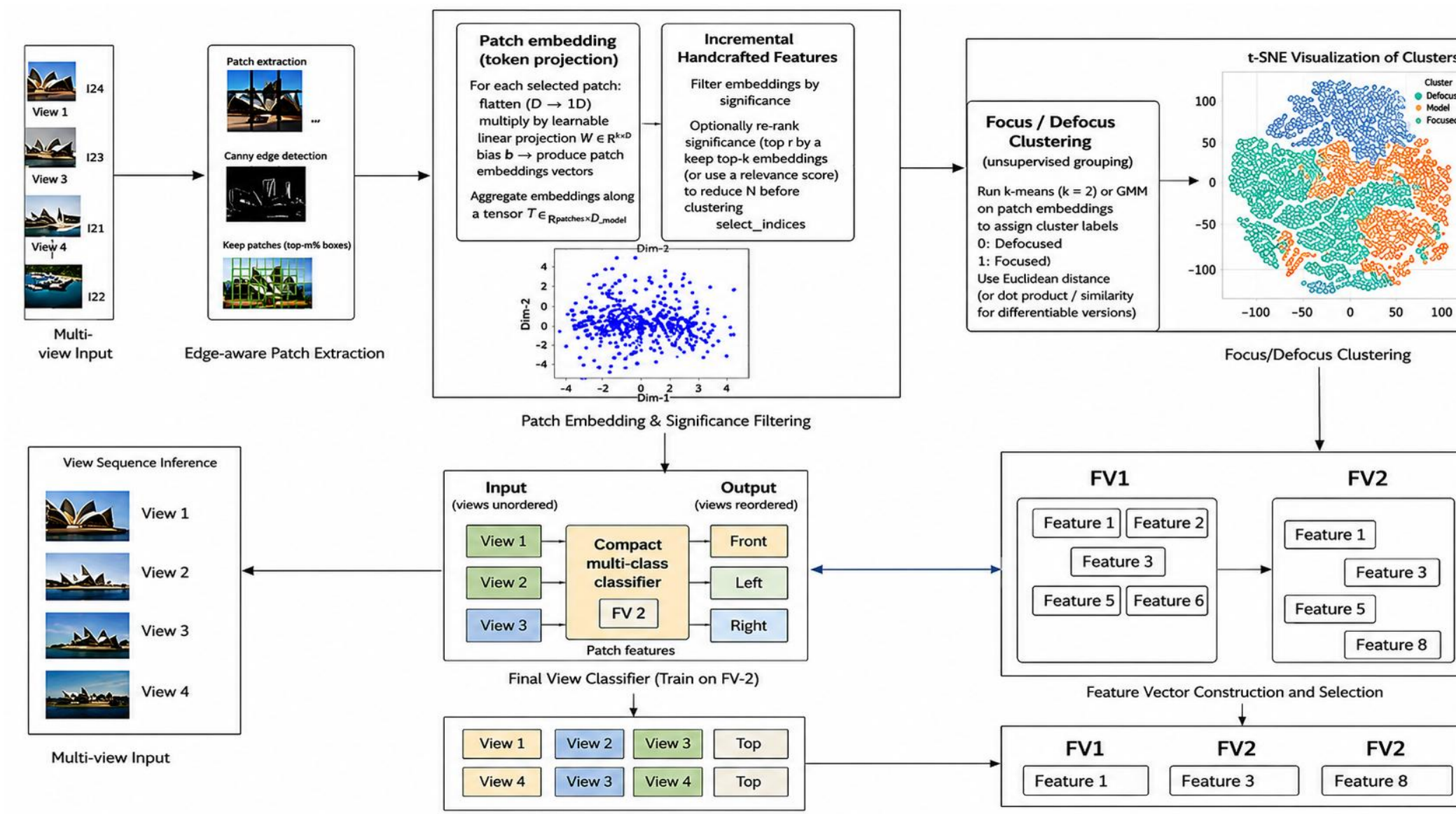
VIEW-4(TOP)

- 150 Scene Sets.
- 600 Images.
- 4 Views.
- Outdoor Environment.

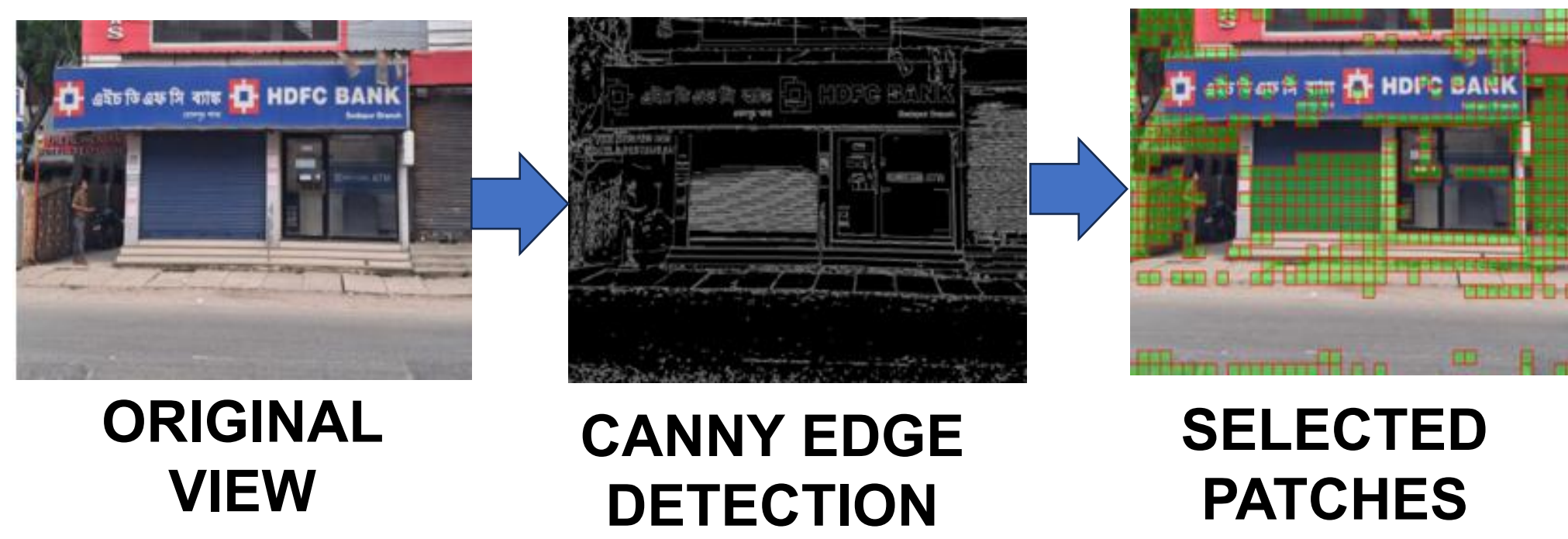
KEY CONTRIBUTIONS

- MSTIS framework
- Edge-aware patch selection
- Focus-Defocus clustering
- FV1 → FV2 feature selection
- Better text detection

PROPOSED FRAMEWORK

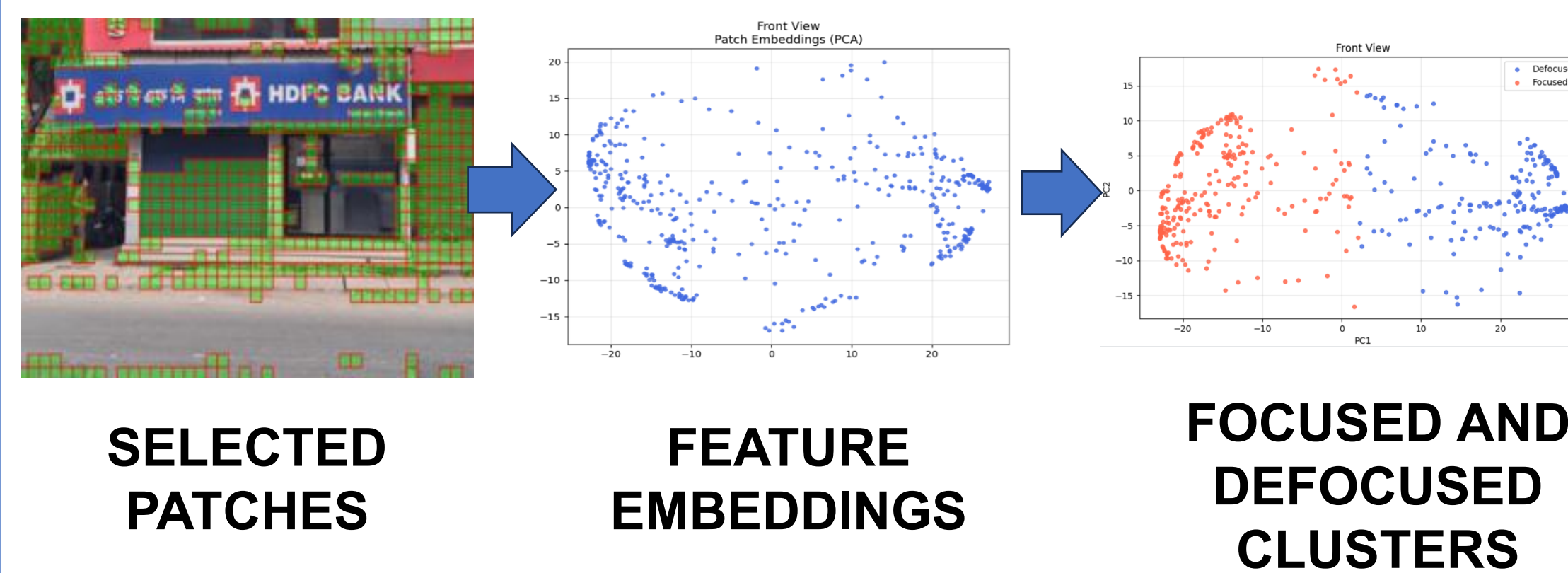


EDGE AWARE PATCH SELECTION



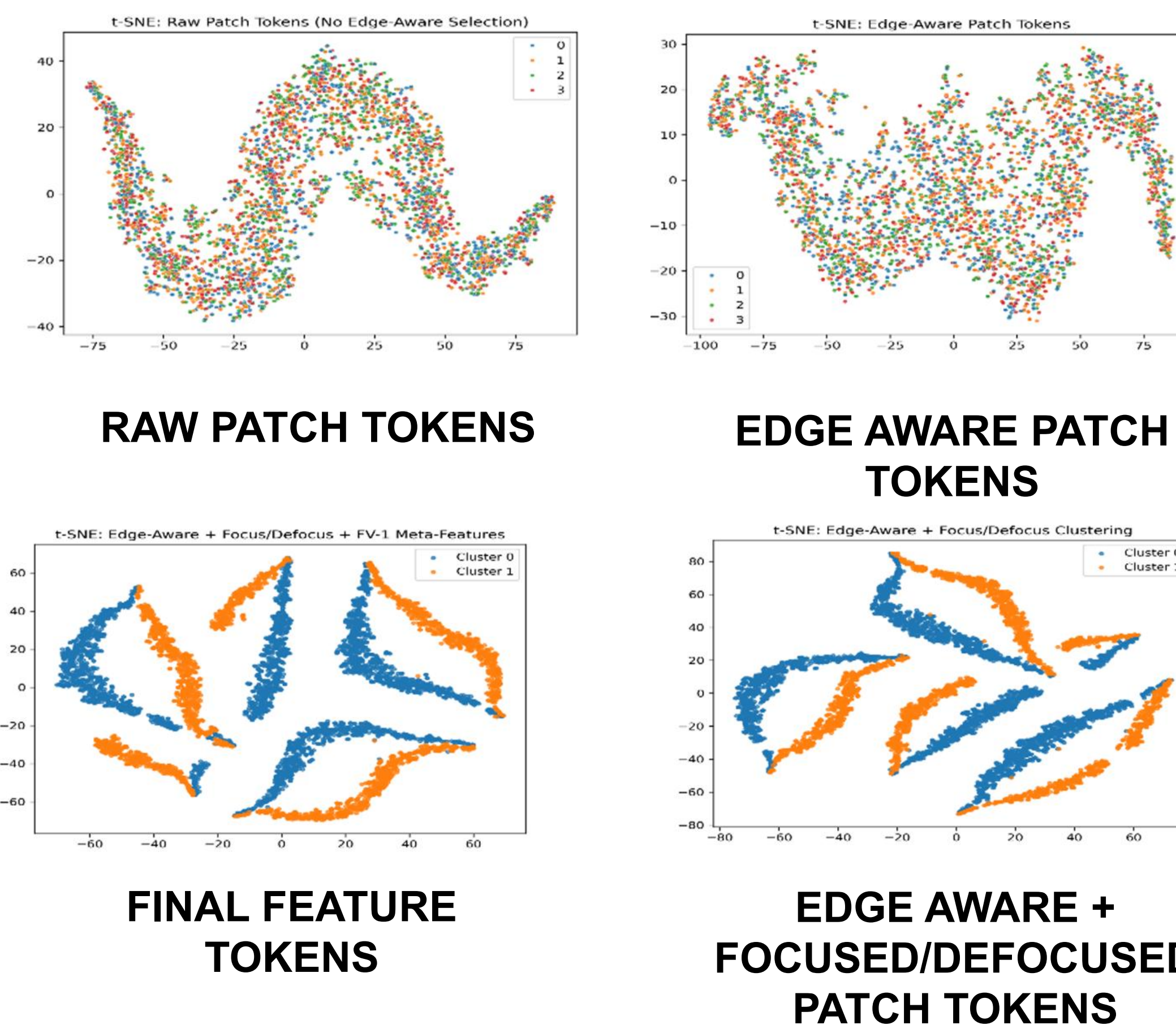
- Detect edges using Canny.
- Compute edge density.
- Retain informative patches.

PATCH EMBEDDING AND FOCUS/DEFOCUS CLUSTERING



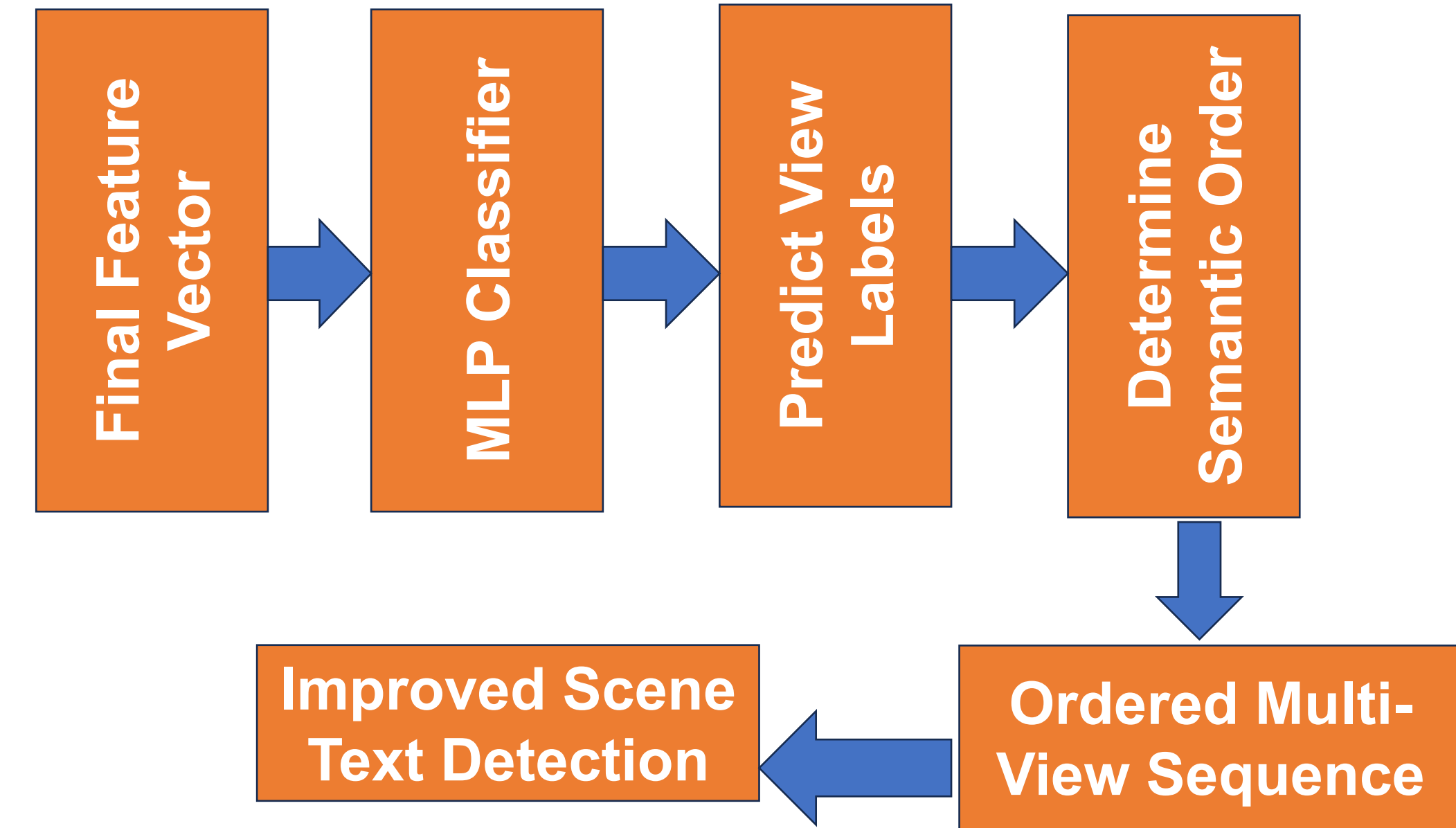
- Swin Transformer extracts patch embeddings.
- K-Means separates focused and defocused patches.
- Focused patches are retained for feature construction.

FEATURE VECTOR CONSTRUCTION AND GREEDY FEATURE SELECTION

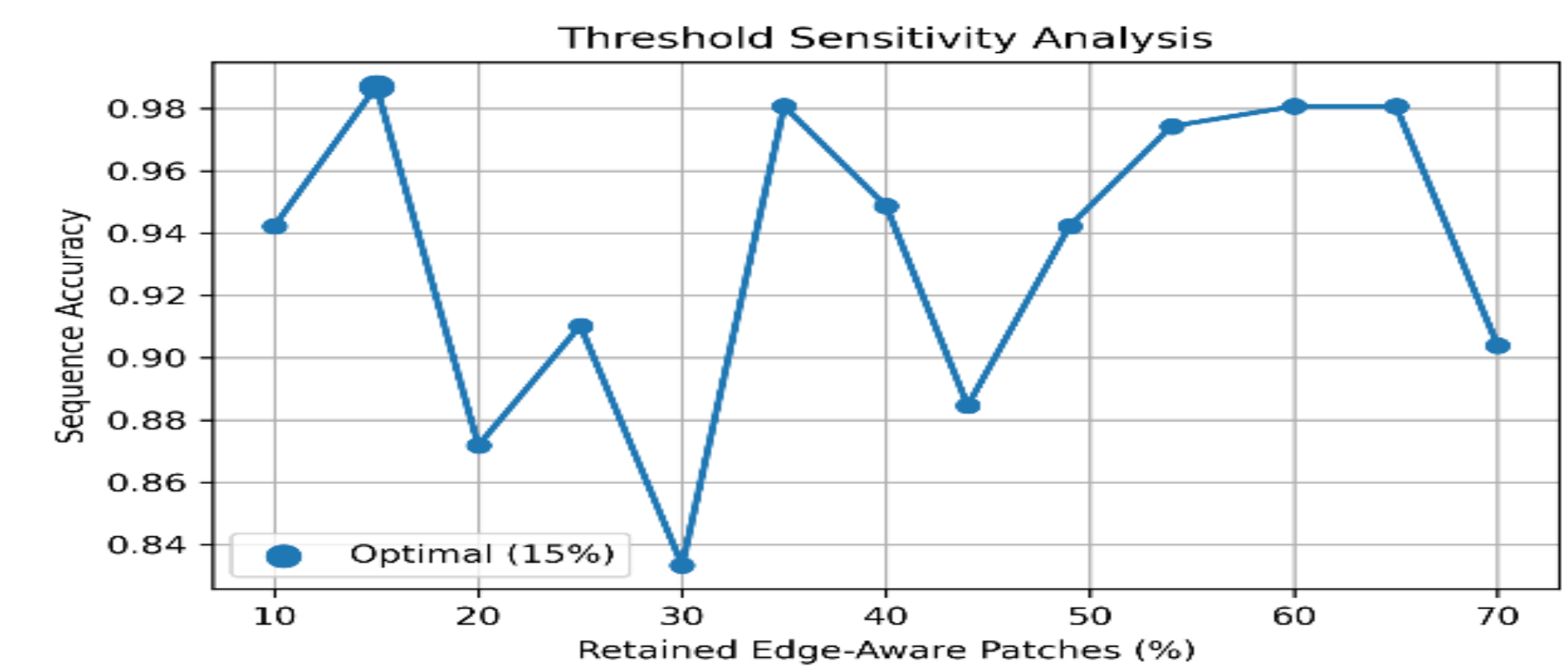


- Initial Feature Vector is constructed.
- Rank and incrementally add candidate features based on overall contribution to the performance.

VIEW-CLASSIFICATION AND SEQUENCE PREDICTION



THRESHOLD SENSITIVITY ANALYSIS



ABLATION STUDY

Configuration	View Classification Accuracy	Sequence Accuracy
Baseline: Raw Patch Tokens Only	53.4	59.0
+ Edge-Aware Patch Selection	63.5	73.4
+ Focus/Defocus Clustering	65.9	73.8
+ FV-1 Feature Construction	79.5	85.7
Proposed MSTIS	93.5	98.5

IMPACT OF VIEW SEQUENCING ON TEXT DETECTION PERFORMANCE

Model	F1-Score(Before Sequenci ng)	F1-Score (After Sequencin g)	Improve- ment
PAN++	90.1	93.4	+3.3
DBNet++	91.3	94.0	+2.7
SwinText	92.0	95.2	+3.2
TextFuseNet	89.8	92.1	+2.3
Average	90.8	93.7	+2.9

QUALITATIVE RESULTS



BEFORE
SEQUENCING

ORIGINAL
DETECTIONS



AFTER
SEQUENCING

MISSED
DETECTIONS

RECOVERED
DETECTIONS

CONCLUSION

- Lightweight MSTIS framework for multi-view sequencing.
- Achieved 93.5% View Classification Accuracy.

- Achieved 98.5% Sequence Accuracy.
- Improved downstream text detection by +2.9 F1-score.